

DeskRadar – Build Guide

An ESP32 with a round 1.28" display shows aircraft in your area live on a radar screen (data from adsb.fi, free, no registration required). These instructions are intended for people who **have never worked with an ESP32 before**.

You can find the matching 3D-printable enclosure on MakerWorld (link from Steffen Moll, see project description).

1. What You'll Need

Hardware

- 1× ESP32-WROOM-32 DevKit (30-pin, with USB-C) – generic board, as shown in the image
- 1× round 1.28" SPI TFT display, 240×240 pixels, **GC9A01** driver IC
- 7 jumper/strand cables (display → ESP32)
- 1× USB-C cable (data, not just a charging cable!) for flashing and power supply
- Optional: 3D-printed enclosure

Software

- [Arduino IDE](#) (version 2.x recommended)
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2. Wiring

Display pin	ESP32 Pin	Meaning
VCC	3V3	Power supply
GND	GND	Ground
SCL	GPIO18	SPI Clock
SDA	GPIO23	SPI Data (MOSI)
CS	GPIO15	Chip Select
DC	GPIO2	Data/Command
RST	GPIO4	Reset
BLK (Backlight)	GPIO21	Backlight (controlled directly in the code)

The built-in **BOOT button** on the ESP32 board (GPIO0) is reused as the Settings button during operation—you don't need to wire anything extra for this.

If your board/display is labeled differently: the pin names on the display board (VCC, GND, SCL, SDA, CS, DC, RST) are standard and should be the same for every GC9A01 display.

3. Set up the Arduino IDE

3.1 Install ESP32 support

1. Open the Arduino IDE → **File** → **Preferences** (Windows/Linux) or **Arduino IDE** → **Settings** (Mac).
2. Under “**Additional Board Manager URLs,**” enter:

```
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
```

3. **Tools** → **Board** → **Board Manager**, then install `esp32`, then install the “**esp32 by Espressif Systems**”.
4. **Tools** → **Board** → Select “ESP32 Dev Module.”

3.2 Install Required Libraries

Tools → **Manage Libraries**, then install the following:

- `TFT_eSPI` (by Bodmer)
- `ArduinoJson` (by Benoit Blanchon, Version 7.x)

(`WiFi`, `Web Server`, `HttpClient`, `Preferences`, `FS`, `SPI` are already included in the ESP32 board package, so you don't need to install anything extra.)

3.3 Configure TFT_eSPI for this display

This is the step where most beginners get stuck—the library doesn't automatically know which display and which pins you're using. This is set via a configuration file named `User_Setup.h`.

1. Find the folder containing the TFT_eSPI library. It's usually located at:

- Windows: `Documents\Arduino\libraries\TFT_eSPI`
- Mac: `~/Documents/Arduino/libraries/TFT_eSPI`

2. Open the file `User Setup Select.h` in this folder using a text editor.

3. Comment out the default line

```
#include <User_Setup.h>
```

— instead, we'll edit `User_Setup.h`.

4. Open `User_Setup.h` and replace the contents with the following settings (or insert them as shown, so that only these driver/pin definitions are active):

```
#define GC9A01_DRIVER

#define TFT_WIDTH 240
#define TFT_HEIGHT 240

#define TFT_MISO -1
#define TFT_MOSI 23
#define TFT_SCLK 18
#define TFT_CS 15
#define TFT_DC 2
#define TFT_RST 4
#define TFT_BL 21
#define TFT_BACKLIGHT ON HIGH

#define LOAD_GLCD
#define LOAD_FONT2
#define LOAD_FONT4

#define SPI_FREQUENCY 40000000
```

These values correspond exactly to the wiring described in Section 2.

4. Upload the code

1. Open `DeskRadar.ino` in the Arduino IDE.
2. Connect the ESP32 to the computer via USB-C.
3. **Tools** → **Port** → Select the appropriate COM/USB port (for some boards, a CP2102/CH340 USB driver must be installed—the IDE will display a message if this is necessary).
4. Click the **upload arrow** (→).
5. If the upload does not start: Hold down the BOOT button on the board while the upload begins, then release it (only necessary for some boards).

After a successful upload, the display boots up and briefly shows `SYSTEM BOOT...`.

5. Initial Setup (Wi-Fi & Location)

When you start the device for the very first time, no Wi-Fi network is saved yet, so the device automatically creates its own access point:

1. The display shows `WIFI FAILED` → `PORTAL ACTIVE` .
2. Use your cell phone or laptop to access `DeskRadar Setup by Steffen Moll` Search for and the Wi-Fi settings menu.
3. Open your browser and `192.168.4.1` open (opens automatically on some phones) enter the address (as shown in the pop-up).
4. Fill out the form:
 - **Device PIN:** `1991`
 - **Select a Wi-Fi network** (from the list) + enter the password
 - **Latitude / Longitude:** Coordinates of your location (e.g., right-click on [Google Maps](#) → Copy coordinates)
 - **Range (KM):** Radius within which aircraft should be displayed
 - Check the boxes for Sweep Animation / Metric System as desired
5. Click **SAVE & REBOOT**. The device will restart and connect to your Wi-Fi. After that, the radar will automatically download flight data from [adsb.fi](#) and display it.

6. Assemble the housing

The 3D-printed housing consists of five parts: **base** (the tray for the ESP32 board), **base cover** (back panel with cutout), **display back panel**, and **display front panel** (shield-shaped half-shells that frame the round display and feature the “Desk Radar” logo), as well as a small **retaining pin**.

This section is based on an analysis of the 3D print files (no actual test assembly yet) and is a **preliminary draft**. Check the assembly steps and let me know if anything doesn't fit.

1. **Insert the ESP32 into the socket.** Place the ESP32 board—which is already wired to the display (see Section 2)—onto the guide rail inside the socket. Align the board so that the USB-C port fits exactly into the side recess of the socket wall.
2. **Insert the retaining pin.** Insert the retaining pin on the side opposite the USB-C port between the board and the socket wall. It gently pushes the board toward the port recess and prevents it from slipping when plugging in or unplugging the charging cable.

3. **Insert the display into the display backplate.** Place the round 1.28" display into the recess of the display backplate.
4. **Place the display back panel onto the base.** Place the display back panel onto the top of the base so that the two holes in the bridge align with the two mounting points on the base.
5. **Place the display front on top and secure it with screws.** Place the display front (with the "Desk Radar" lettering) on top from the front so that the display is clamped between the front and back. Secure it with two small screws through both parts into the mounting points on the base—this secures the display and the top half of the housing in one step.
6. **Place the base cover on top.** Slide the base cover onto the base from the back; the recess in the cover encloses the tab on the back of the display. The cover seals the housing at the back.

7. Change settings later

Press and hold the **BOOT button** on the ESP32 (or the button on the housing, if installed) **for 3 seconds** → The device will restart and reopen the setup portal as in Step 5, with all previous values pre-filled.

8. Troubleshooting

Problem	Solution
Display remains black	Check the wiring (especially CS = GPIO15, not GPIO5, which is often mentioned in tutorials <code>GPIO5</code>), check <code>User_Setup.h</code>
Upload fails / Port missing	Install USB driver (CP2102/CH340), try a different USB cable (data cable, not just a charging cable)
"WIFI FAILED" on every boot	Wi-Fi password saved incorrectly → Hold the Settings button for 3 seconds, then set it up again
No aircraft visible	Increase the range (km), check if there is actually air traffic nearby, check the ESP32's internet connection